

Association of Fitzpatrick Skin Type with Metastatic Risk from Uveal Melanoma in 854 Consecutive Patients at a Single Center

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Article

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Abstract

Objective: To assess the association of skin color using Fitzpatrick Skin Type (FST) with metastatic risk of uveal melanoma.

Subjects: 854 consecutive patients with uveal melanoma and documented FST

Methods: Retrospective detailed review of patient charts was performed for FST (type I- white, II-fair, III-average, IV-light brown, V-brown, VI-black), clinical details of the patient and the uveal melanoma, tumor cytogenetic classification according to The Cancer Genome Atlas (TCGA), and outcome of melanoma-related metastasis and death.

Results: The FST classification was type I (n=97 patients), type II (n=665), type III (n=79), type IV (n=11), type V (n=2), type VI (n=0). A comparison of patient FST (type I vs. II vs. III-V) revealed significant differences in mean age at presentation (64.1 vs. 58.5 vs. 49.8 years, $p<0.001$), race white (100% vs. 98% vs. 75%, $p<0.001$), presence of ocular melanocytosis (3% vs. 3% vs. 10%, $p=0.01$), visual acuity $<20/200$ at presentation (6% vs. 7% vs. 13%, $p=0.03$), genetic results showing TCGA group B tumors (11% vs. 14% vs. 26%, $p=0.01$) or TCGA group D tumors (22% vs. 11% vs. 9%, $p=0.01$), 10-year incidence of melanoma-related metastasis (25% vs. 15% vs. 14%, $p=0.02$) and 10-year incidence of melanoma-related death (9% vs. 3% vs. 4%, $p=0.04$). FST was a significant predictor of melanoma-related metastasis ($p=0.02$, Hazard ratio 2.3).

Conclusions: Fitzpatrick skin type may be a predictor of melanoma-related metastasis, with metastasis and TCGA Group D tumors being more common in patients with FST I.

Introduction

The Fitzpatrick Skin Type (FST) classification scheme is used to classify skin pigmentation on a scale from I (fair skin) to VI (dark skin).^{1,2} This classification was established by Fitzpatrick, a dermatologist, in 1975 and was initially used to judge the appropriate use of ultraviolet light for the treatment of psoriasis.^{1,2} The risk of developing cutaneous melanoma, basal cell carcinoma and squamous cell carcinoma has been shown to inversely correlate with FST.³⁻⁹ While patients with lower FST may be at increased risk of melanomagenesis, cutaneous melanoma tends to present at a more advanced stage and is associated with worse outcomes in patients with higher FST.¹⁰⁻¹⁷ Conjunctival melanoma has been shown to be more common in patients with lower FST, but FST is not associated with different outcomes in terms of vision loss, tumor recurrence, exenteration, metastasis or death.¹⁸

Several authors have established that uveal melanoma is more common in fair-skinned, blue- or green-eyed patients than those with darker iris color.¹⁹⁻²⁵ Chromosome 3 and 8 aberrations have been well described as a factor predictive of uveal melanoma metastatic risk²⁶⁻²⁹. The Cancer Genome Atlas (TCGA) classification groups uveal melanomas according to its metastatic risk based on chromosomal abnormalities with Group A (disomy 3 and 8) having the lowest risk of metastasis and subsequent

classes (Group B (disomy 3 and 8q gain), Group C (monosomy 3 and 8q gain) and Group D (monosomy 3 and multiple 8q gains)) having progressively increased risk.^{26,27,29} Chromosome 3 and 8 aberrations have perhaps the greatest predictive influence in patients with light irides compared to those with darker irides.³⁰

While there have been assessments of uveal melanoma outcomes based on patient race, there have been no previous studies to exploring outcomes based on a standardized classification of skin color.³¹⁻³⁴ Herein, we review FST in patients with uveal melanoma to identify if there are differences in clinical features, genetic results, or outcomes.

Methods

The medical records of the Ocular Oncology Service at Wills Eye Hospital, Philadelphia, USA were retrospectively reviewed for patients with a diagnosis of uveal melanoma, between 2006 and 2021 who underwent genetic evaluation and assessment for TCGA classification and had documented recording of FST (Figure 1). This study has been reviewed by the Wills Eye Hospital Institutional Review Board (IRB) and has been rendered exempt under category 4 (secondary research for which consent is not required). This research involved retrospective information collection and analysis without identifiable private information or identifiable biospecimens. It adhered to the tenets of the Declaration of Helsinki and complied with the Health Insurance Portability and Accountability Act. Informed consent was obtained from all patients for inclusion in the study and for inclusion of images of their faces when relevant. All authors have no conflicts of interest to disclose.

The diagnosis of uveal melanoma was established on the Ocular Oncology Service by an ocular oncologist using indirect ophthalmoscopy and multimodal ophthalmic imaging, including wide-angle fundus photography, ultrasonography, fundus autofluorescence, optical coherence tomography (OCT), fluorescein angiography, and indocyanine green angiography when necessary. Patients were classified according to the Fitzpatrick Skin Type (FST) classification (Types I–VI) at the time of their initial diagnosis or retrospectively according to external facial photographs (Figure 1). This classification focuses on skin color described as Type I (white), Type II (fair), Type III (average), Type IV (light-brown), Type V (brown), and Type VI (black), with increasing suntan and decreasing sunburn potential, eye color is not included in the classification system.¹⁻² All patients included in the study had undergone genetic testing by fine-needle aspiration biopsy (FNAB) or open biopsy of the tumor, as previously described in the literature²⁹, with results on the status of chromosomes 3, 6, and 8 allowing classification according to TCGA as groups A, B, C or D.³⁵

Data was collected from each patient chart regarding patient demographics, clinical features of the patient and tumor, genetic features of the tumor, and outcomes. Demographic data included patient age, race, sex and affected eye. All patients attending the clinic have their race recorded at their initial visit by the physician after discussion with the patient. Clinical data at initial examination included best corrected visual acuity (VA), presence of ocular melanocytosis or heterochromia, tumor epicenter location by tissue

(choroid, ciliary body, iris), distance from tumor edge to optic disc and foveola (millimeters [mm]), and tumor basal diameter by fundus photography (mm) and thickness by ultrasonography (mm). Tumor cytogenetic data using TCGA classification (TCGA groups A, B, C, and D) were noted as well as data on treatment modality (plaque radiotherapy, tumor resection (partial lamellar sclerouvectomy), enucleation).

Outcomes data included date last seen visual acuity, change in visual acuity over time, melanoma recurrence and treatment, and melanoma-related metastasis and death. Information on metastasis and death was gathered through history from the patient/family and correspondence from physicians. Metastasis was further stratified based on the rate of metastasis to the liver, lung, and other systemic locations.

Statistical Analysis

Statistical analysis was performed using the R Project for Statistical Computing (version 4.0.2; The R Foundation, Vienna, Austria). FST categories III, IV and V were grouped together for all descriptive statistics and analyses due to low numbers. Continuous variables were expressed as mean (median, range) and categorical variables were expressed as patient number (percentage). After performing the Shapiro-Wilk test to assess for normality, comparisons between the three FST classification categories (Types I vs. II vs III, IV, and V) were made using the one-way ANOVA test for continuous variables with normal distribution and Kruskal-Wallis H test for continuous variables without normal distribution. Comparisons for categorical variables were performed using the Chi-square test or Fisher exact test when indicated. In order to calculate patient percentages and test for independence among FST subgroups for TCGA classification, a binary variable was created for each TCGA classification (A-D) and Chi-squared tests were performed. Post-hoc pairwise comparisons were done using Dunn's tests for continuous variables with non-normal distributions and t-tests with pooled standard deviation for categorical variables. All p -values for the pairwise comparisons were adjusted using the Bonferroni method to ensure conservative estimates. Kaplan-Meier analysis was performed to determine cumulative probability of outcomes, including metastasis and death. The log-rank test was performed to assess differences in survival distribution between the Fitzpatrick categories. Hazard ratios and 95% confidence intervals for melanoma-related metastasis and melanoma-related death were calculated for FST and relevant covariates using the Cox proportional hazards model having confirmed the proportional hazard assumption using the Schoenfeld test. Variable selection for the Cox regression analysis was performed via univariate regression followed by stepwise logistic regression. A p -value <0.05 was considered statistically significant.

Results

There were 854 patients diagnosed with uveal melanoma between 2006 and 2021 who had FST assessment and tumor DNA cytogenetic testing, classified by TCGA. The FST classification was type I (n=97 patients), type II (n=665), type III (n=79), type IV (n=11), type V (n=2), type VI (n=0).

The demographic features of the patients are listed in Table 1. A comparison of the FST (type I vs. II vs. III-V) revealed differences in mean age (64.1 vs. 58.5 vs. 49.8 years, $p<0.001$) and race white (100% vs. 98% vs. 75%, $p<0.001$). There were no differences in sex and involved eye.

The clinical and tumor features at presentation are listed in Table 2. A comparison of the FST (type I vs. II vs. III-V) revealed differences in percentage of patients with ocular melanocytosis (2% vs. 3% vs. 9%, $p=0.01$), initial visual acuity $<20/200$ (6% vs. 7% vs. 13%, $p=0.03$), mean largest basal diameter (13.2 vs. 12.0 vs. 13.3 mm, $p=0.003$), mean tumor thickness (6.0 vs. 5.6 vs. 6.7 mm, $p=0.03$), percentage of patients with TCGA group B tumors (11% vs. 14% vs. 26%, $p=0.01$) and percentage of patients with TCGA group D tumors (22% vs. 11% vs. 9%, $p=0.01$). There were no differences in heterochromia, distance to optic disc, distance to foveola and tumor epicenter.

A pairwise comparison for largest basal diameter demonstrated that both the FST I vs FST II (13.2 vs 12.0 mm) and the FST II vs FST III-V (12.0 vs 13.3 mm) comparisons were significant ($p=0.04$ and $p=0.02$ respectively). For tumor thickness, the FST II vs FST III-IV (5.6 vs 6.7 mm) comparison was significant ($p=0.03$). For initial visual acuity, the FST II vs FST III-V comparison was significant (75% vs 62% with 20/50 or better visual acuity, $p=0.02$). Pairwise comparisons for percentage of TCGA group B patients demonstrated the FST I vs FST III-V (11% vs 26%) and the FST II vs FST III-V (14% vs 26%) comparisons were significant ($p=0.01$ for both comparisons). For percentage of patients with TCGA group D tumor cytogenetics, the FST I vs FST II (22% vs 11%) and the FST I vs FST III-V (22% vs 9%) comparisons were both significant ($p=0.01$ and $p=0.02$, respectively).

The primary treatment modality is listed in Table 3. Plaque radiotherapy was performed in 793 (93%), enucleation in 55 (7%) and partial lamellar sclerouvectomy (PLSU) in 2 ($<1\%$) patients. There was no difference in treatment modality between FST groups.

Outcomes following treatment are listed in Table 4. The mean follow-up duration was 44 months (median 34, range 0.1-210 months). A comparison of the FST (type I vs. II vs. III-V) revealed differences in 10-year cumulative incidence of melanoma-related metastasis (25% vs 15% vs 14%, $p=0.02$) (table 4). Kaplan-Meier survival curves demonstrating significant differences between FST groups for melanoma-related metastasis ($p=0.04$) and non-significant differences for melanoma-related death ($p=0.12$) are shown in Figure 2. The Kaplan-Meier (2-year/5-year/10-year) rate of metastasis was 17%/33%/not available for FST I, 9%/20%/30% for FST II and 7%/27%/34% for FST III-V. There were no differences in visual acuity at date last seen, rates of visual acuity loss ≥ 3 Snellen lines, local recurrence rate, treatment modality of local recurrence, 2-year and 5-year melanoma-related metastasis rate, location of metastasis, and 2-year, 5-year and 10-year melanoma-related death rates (table 4).

Cox Proportional Hazard results for melanoma-related metastasis and melanoma-related death are shown in Table 5. The factors best predictive of melanoma-related metastasis included TCGA (Group A vs Group D, $p<0.001$; Group A vs Group C, $p<0.001$; Group A vs Group B, $p=0.02$), largest basal diameter ($p<0.001$) and FST (FST I vs FST III-V, $p=0.02$). Furthermore, TCGA (Group A vs Group D, $p<0.001$; Group A vs Group C, $p=0.046$), largest basal diameter ($p<0.001$) and local recurrence ($p=0.02$) were also predictive

of melanoma-related death. The rate of melanoma related metastasis in TCGA group D patients was 18.6 times greater than that of those in TCGA group A.

Discussion

Uveal melanoma clinical presentation and outcomes have been shown to vary by race.³¹⁻³⁴ In a study of race and uveal melanoma in 8100 patients, Caucasian patients were noted to present at an older age than Hispanic, Asian and African American patients (58, 48, 44 and 52 years; $p<0.001$), with smaller tumor basal dimension (11, 12, 12, 13 mm, $p<0.001$) and smaller tumor thickness (5.4, 7.1, 6.5, 7.5 mm, $p<0.001$).³³ Although no significant difference was demonstrated in the 8100 patients in terms of melanoma-related metastasis or death based on race, the relative risk for metastasis, compared with Caucasians, was 0.31 for African Americans, 0.73 for Hispanics and 1.42 for Asians.³³ An analysis of the Surveillance, Epidemiology and End Results (SEER) registries from 2004 to 2014 in 4130 patients with uveal melanoma revealed that non-white patients (compared to white patients) were more likely to be diagnosed with uveal melanoma at a younger age (<55 years old) (48% vs 33%, $p<0.001$) and receive management of primary enucleation (OR, 1.45 95% CI, 1.12-1.88).³² Disease-specific survival did not vary significantly by race in that SEER analysis.³² Using race as a variable can be problematic given that race is a social category and self-identified and not a biological category and race can change over time and vary across societies.^{36,37} FST could be a more objective alternative to race to estimate a patient's skin tone and evaluate whether skin tone has an influence on uveal melanoma features and outcomes.

Cutaneous melanoma has been shown in multiple studies to present at a more advanced stage and to have worse outcomes in patients with darker FST.¹⁰⁻¹⁷ Determining whether this is due to skin color or socio-demographic status has been problematic.¹¹ A recent study of 135 patients with cutaneous melanoma from Malta showed that patients with FST I and II were more likely to have invasive disease than those with FST III and IV.³ Conjunctival melanoma incidence has been increasing over time in the white populations of Europe and the United States.^{38,39} In Finland, the annual age-adjusted incidence increased from 0.4 to 0.8 per million white inhabitants between 1967 and 2000.³⁸ In the United States, the age-adjusted incidence rate for white men increased 295% between 1973 and 1999.³⁹ The incidence and outcomes of conjunctival melanoma based on race have not been well studied because of the rarity of this malignancy. However, a recent study of conjunctival melanoma outcomes based on FST in 540 patients confirmed that this tumor is far more common in patients with lower FST, but FST was not associated with various outcomes including vision loss, tumor recurrence, exenteration metastasis or death.¹⁸

In this report, we describe the relationship of FST and uveal melanoma clinical features and outcomes. We have shown that patients with darker skin tone are less likely to be classified racially as white, present at a younger age, and are more likely to have ocular melanocytosis and worse visual acuity at presentation. Patients with darker skin tone are less likely to have high risk tumor cytogenetics based on TCGA classification. Furthermore, we have documented that FST is a predictor of melanoma-related

metastasis as well as TCGA group and that patients with darker skin tone may be less likely to develop melanoma-related metastasis than those with lighter skin tone.

It is well established that light iris color is associated with increased risk of iris melanocytic lesions and uveal melanoma.^{20,40} The association between iris color and uveal melanoma could be independent of skin tone or it could actually represent an accurate surrogate for FST. A recent Dutch study of 412 patients with uveal melanoma and 5951 control patients showed that patients with light iris colors of green/hazel (Odds Ratio=3.64) or blue/grey (Odds ratio=1.38) demonstrated higher crude risk of uveal melanoma than those with brown irides.¹⁹ Furthermore, the biology of uveal melanoma in eyes with different iris colors can differ, with chromosome 3 and 8 aberrations imparting a greater influence with reduction in patient survival with light irides compared to brown irides.³⁰

There are several limitations to this study, beginning with its retrospective nature. Despite the retrospective nature of the data collection, the FST had been charted at date first seen prospectively over 15 years. The ideal method of FST collection has been described with observation of skin tone in non-sun-exposed skin, such as the buttocks or inner arm⁶. As this is often not practical in an ophthalmology clinic, facial skin, particularly in the sub-brow, minimally-sun-exposed area, was assessed as a surrogate which could be less accurate than purely non-sun-exposed skin. Another limitation is that outcomes for metastasis and death were per report by the patient, family, or physician, following systemic monitoring on a twice-yearly basis with liver and lung imaging. Data collection ended with the patient last visit or office correspondence to avoid missing metastatic or death events. However, there could have been some overlooked metastatic events. Patients, families, and physicians may have been less likely to report death events than metastatic events which could explain why no firm conclusions can be drawn on this outcome's relationship with FST. Due to a limited number of patients being followed-up for 6 years or greater, the 10-year outcome data needs to be interpreted with caution.

In conclusion, we present novel observations on the relationship between skin tone using Fitzpatrick Skin Type (FST) and clinical features, genetic outcomes, and metastatic risk of uveal melanoma. In this large cohort of 854 consecutive patients, we have shown that FST I patients are at increased risk for uveal melanoma metastasis and high-risk TCGA tumor cytogenetic classification.

Declarations

Conflict of interest

No conflict of interest exists for any of the authors in relation to this work.

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Tables

Table 1. Uveal melanoma outcomes based on the Fitzpatrick Skin Type (FST) classification in 854 consecutive patients: Demographics.

Demographics	Fitzpatrick Skin Type I (n= 97 patients) [n (%)]	Fitzpatrick Skin Type II (n= 665 patients) [n (%)]	Fitzpatrick Skin Type III, IV & V (n= 92 patients) [n (%)]	<i>p</i> values	Total (n= 854 patients) [n (%)]
Patient Age (years)					
Mean (median, range)	64.1 (66.0, 27.0-94.0)	58.5 (60.0, 10.0-92.0)	49.8 (50.5, 17.0-90.0)	<0.001	58.2 (60.0, 10.0-94.0)
Sex					
Male	50 (52)	344 (52)	52 (57)	0.68	446 (52)
Female	47 (48)	321 (48)	40 (43)		408 (48)
Race					
White	97 (100)	652 (98)	69 (75)	<0.001	818 (96)
Black	0 (0)	0 (0)	2 (2)		2 (<1)
Hispanic	0 (0)	9 (1)	9 (10)		18 (2)
Asian	0 (0)	1 (<1)	5 (5)		6 (1)
Other	0 (0)	0 (0)	5 (5)		5 (1)
Unknown	0 (0)	3 (1)	2 (2)		5 (1)
Involved Eye					
Right	55 (57)	359 (54)	43 (47)	0.35	457 (54)
Left	42 (43)	306 (46)	49 (53)		397 (46)

Bold values indicate statistical significance.

Table 2. Uveal melanoma outcomes based on the Fitzpatrick Skin Type (FST) classification in 854 consecutive patients: Clinical and tumor features at presentation.

Clinical Features	Fitzpatrick Skin Type I (n= 97 patients) [n (%)]	Fitzpatrick Skin Type II (n= 665 patients) [n (%)]	Fitzpatrick Skin Type III, IV, & V (n= 92 patients) [n (%)]	<i>p</i> values	Total (n= 854 patients) [n (%)]
Ocular Melanocytosis					
Total number of patients	2 (2)	18 (3)	8 (9)	0.01	28 (3)
Heterochromia					
Total number of patients	1 (1)	13 (2)	3 (3)	0.10	17 (2)
Visual Acuity					
20/20-20/50	66 (68)	501 (75)	57 (62)	0.03	624 (73)
20/60-20/200	25 (26)	115 (17)	23 (25)		163 (19)
20/400-NLP	6 (6)	49 (7)	12 (13)		67 (8)
Distance to Optic Disc (mm)					
Mean (median, range)	4.9 (4.0, 0.0-15.0)	4.5 (3.5, 0.0-20.0)	4.9 (4.0, 0.0-19.0)	0.40	4.6 (4.0, 0.0-20.0)
Distance to Foveola (mm)					
Mean (median, range)	4.7 (4.0, 0.0-15.0)	4.3 (3.0, 0.0-18.4)	4.3 (3.0, 0.0-17.0)	0.29	4.4 (3.0, 0.0-18.4)
Largest Basal Diameter (mm)					
Mean (median, range)	13.2 (14.0, 5.0-24.0)	12.0 (12.0, 1.0-24.0)	13.3 (14.0, 3.5-22.0)	0.003	12.3 (12.0, 1.0-24.0)
Tumor Thickness DFS (mm)					
Mean (median, range)	6.0 (5.1, 1.0-20.4)	5.6 (4.7, 1.0-19.8)	6.7 (5.8, 0.7-16.0)	0.03	5.8 (4.9, 0.7-20.4)
Tumor Epicenter					
Choroid	92 (95)	603 (91)	77 (84)	0.19	772 (90)
Ciliary Body	4 (4)	49 (7)	11 (12)		64 (8)

Iris	1 (1)	11 (2)	4 (4)		16 (2)
The Cancer Genome Atlas					
Group A	46 (47)	323 (49)	38 (41)	0.42	407 (48)
Group B	11 (11)	92 (14)	24 (26)	0.01	127 (15)
Group C	19 (20)	178 (27)	22 (24)	0.29	219 (26)
Group D	21 (22)	72 (11)	8 (9)	0.01	101 (12)

Bold values indicate statistical significance.

Table 3. Uveal melanoma outcomes based on the Fitzpatrick Skin Type (FST) classification in 854 consecutive patients: Treatment.

Treatment/Management	Fitzpatrick Skin Type I (n= 97 patients) [n (%)]	Fitzpatrick Skin Type II (n= 661 patients‡) [n (%)]	Fitzpatrick Skin Type III, IV, & V (n= 92 patients) [n (%)]	p values	Total (n= 850 patients) [n (%)]
Treatment					
Plaque radiotherapy	93 (96)	614 (93)	86 (93)	0.63	793 (93)
Enucleation	4 (4)	45 (7)	6 (7)	0.76	55 (7)
Partial lamellar sclerouvectomy	0 (0)	2 (<1)	0 (0)	>0.99	2 (<1)

Bold values indicate statistical significance.

‡ lost to follow-up after fine needle aspiration biopsy n=2, treated with stereotactic radiotherapy n=1, no treatment recommended n=1

Ancillary treatments to plaque included transpupillary thermotherapy (TTT) and photodynamic therapy (PDT)

Table 4. Uveal melanoma outcomes based on the Fitzpatrick Skin Type (FST) classification in 854 consecutive patients: Outcomes.

Outcomes	Fitzpatrick Skin Type I (n= 97 patients) [n (%)]	Fitzpatrick Skin Type II (n= 665 patients) [n (%)]	Fitzpatrick Skin Type III, IV, V (n= 92 patients) [n (%)]	p values	Total (n= 854 patients) [n (%)]
Follow-up	n=95	n=650	n=90		N=832
Duration (months) Mean (median, range)	39.9 (34.6, 1.2-115.3)	45.1 (34.6, 0.1-209.8)	43.4 (29.6, 0.1- 153.5)	0.50	44.3 (33.9, 0.1- 209.8)
Snellen Visual Acuity at Date Last Seen	n=91	n=614	n=81		N=786
20/20 – 20/40	30 (33)	222 (36)	25 (31)	0.87	277 (35)
20/50 – 20/150	27 (30)	168 (27)	23 (28)		218 (28)
20/200 or worse	34 (37)	224 (36)	33 (41)		291 (37)
Visual acuity (Snellen) Mean (median, range)	20/80 (20/80, 20/20-NLP)	20/80 (20/70, 20/20-NLP)	20/80 (20/60, 20/20-LP)	0.89	20/80 (20/70, 20/20- NLP)
Visual acuity (LogMAR) Mean (median, range)	0.9 (0.6, 0.0- 3.0)	0.9 (0.5, 0.0- 3.0)	0.9 (0.5, 0.0- 2.7)		0.9 (0.5, 0.0-3.0)
Visual Acuity Loss					
Vision Loss ≥3 Snellen lines	57 (59)	371 (56)	46 (50)	0.27	474 (56)
Recurrence melanoma					
After Primary Treatment	3 (3)	20 (3)	2 (2)	1.00	25 (3)
Treatment of Melanoma Recurrence					
Plaque	1 (50)	4 (21)	0 (0)	0.59	5 (22)
Enucleation	0 (0)	9 (47)	2 (100)		11 (48)
TTT	1 (50)	4 (21)	0 (0)		5 (22)
Stereotactic radiotherapy	0 (0)	1 (5)	0 (0)		1 (4)

Other	0 (0)	1 (5)	0 (0)		1 (4)
Melanoma-Related Metastasis					
Total number of patients	25 (26)	106 (16)	14 (15)	0.06	145 (17)
2-year incidence	13 (13)	51 (8)	5 (5)	0.53	69 (8)
5-year incidence	21 (22)	89 (13)	12 (13)	0.21	122 (14)
10-year incidence	24 (25)	103 (15)	13 (14)	0.02	140 (16)
Melanoma Metastasis Systemic Locations‡					
Liver	24 (25)	102 (15)	12 (13)	0.05	138 (16)
Lung	4 (4)	23 (3)	2 (2)	0.75	39 (5)
Brain	2 (2)	3 (0.5)	0 (0)	0.19	29 (3)
Bone	9 (9)	28 (4)	2 (2)	0.06	5 (1)
Others**	6 (6)	18 (3)	4 (4)	0.15	28 (3)
1 Site of Metastasis	13 (13)	62 (9)	11 (12)	0.33	86 (10)
2 Sites of Metastasis	6 (6)	25 (4)	1 (1)	0.19	32 (4)
≥3 Sites of Metastasis	6 (6)	19 (3)	2 (2)	0.21	27 (3)
Melanoma-Related Death					
Total number of patients	9 (9)	23 (3)	4 (4)	0.04	36 (4)
2-year incidence	2 (2)	6 (1)	1 (1)	0.29	9 (1)
5-year incidence	7 (7)	18 (3)	2 (2)	0.07	27 (3)
10-year incidence	7 (7)	22 (3)	4 (4)	0.15	33 (4)

Bold values indicate statistical significance.

‡ Multiple patients had metastasis to more than one systemic location.

** Other Melanoma Metastasis Systemic Locations included pancreas, omentum, scalp, mesentery, skin, peritoneum, spleen, breast, cerebellum, pelvis, gallbladder, paraspinal muscles, and kidney

LP= light perception, NLP= no light perception

Table 5. Uveal melanoma outcomes based on the Fitzpatrick Skin Type (FST) classification in 854 consecutive patients: Cox Proportional Hazard Results.

Relevant Covariates‡	Melanoma-Related Metastasis (n= 145 patients)	<i>p</i> values	Melanoma-Related Death (n= 36 patients)	<i>p</i> values
	Hazard Ratio (95%CI)		Hazard Ratio (95%CI)	
FST				
FST I vs. FST III-V	2.31 (1.17, 4.56)	0.02	—	—
FST II vs. FST III-V	1.64 (0.92, 2.93)	0.10	—	—
FST I vs. FST II	1.41 (0.90, 2.21)	0.14		
TCGA				
Group A vs. Group D	18.6 (10.1, 34.3)	<0.001	19.9 (6.32, 62.8)	<0.001
Group A vs. Group C	7.29 (4.00, 13.3)	<0.001	3.44 (1.02, 11.6)	0.046
Group A vs. Group B	2.42 (1.13, 5.19)	0.02	0.57 (0.06, 5.15)	0.60
Largest Basal Diameter	1.19 (1.13, 1.25)	<0.001	1.20 (1.09, 1.33)	<0.001
Local Recurrence	—	—	3.69 (1.20, 11.3)	0.02

Bold values indicate statistical significance.

‡ Relevant covariates for the outcomes of Melanoma-Related Metastasis and Melanoma-Related Death were selected via stepwise logistic regression.

Figures

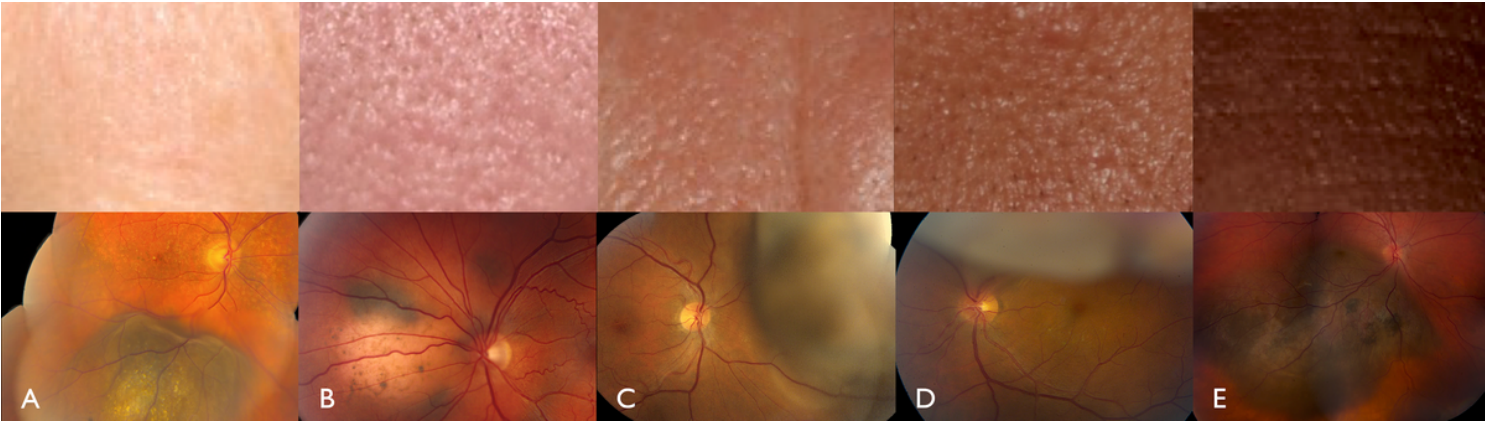


Figure 1

Fitzpatrick skin type (skin type I (A), type II (B), type III (C), type IV (D) and type V(E)) in patients with uveal melanoma and the corresponding tumors on color fundus photography.

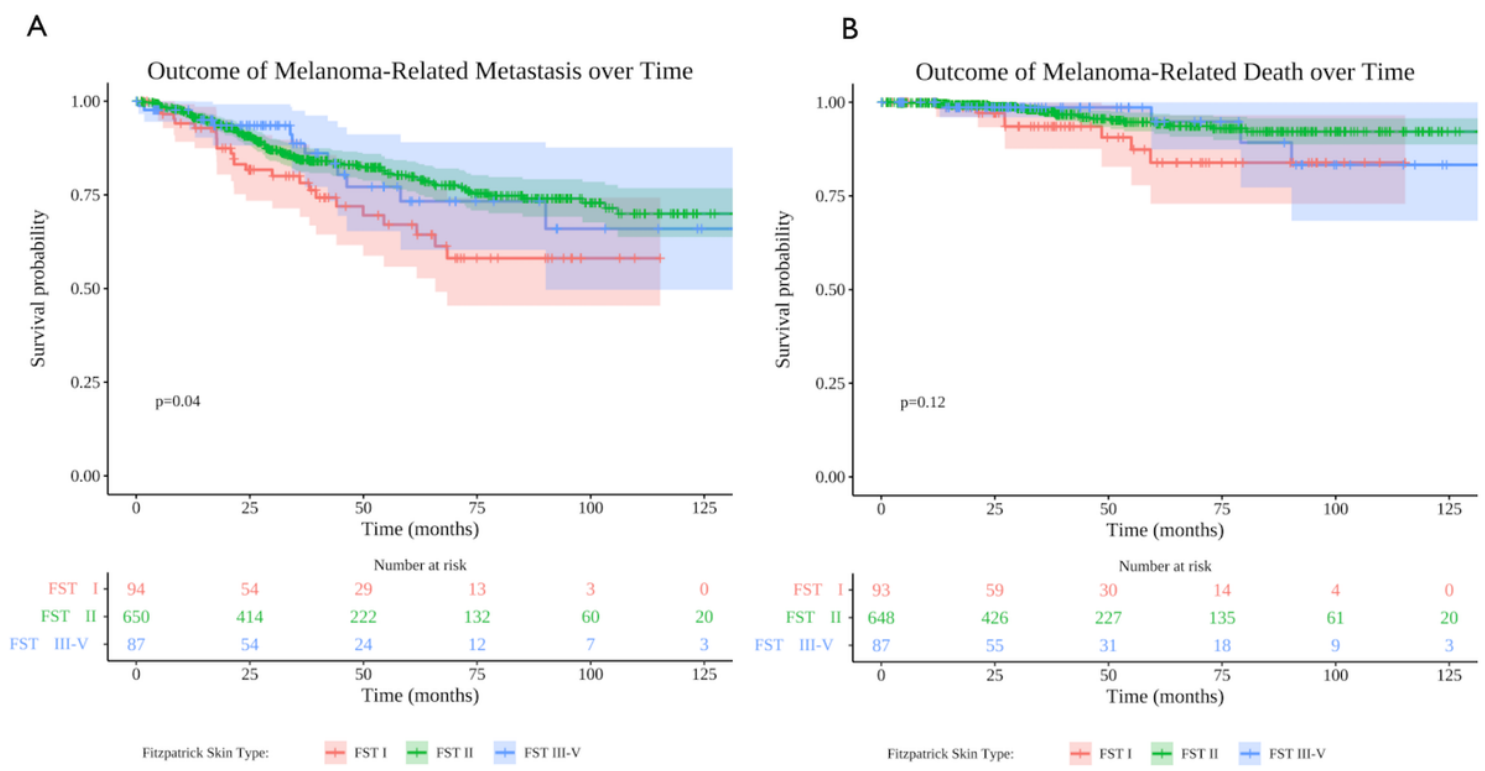


Figure 2

Kaplan-Meier analysis of uveal melanoma-related metastasis (Figure 2A) and death (Figure 2B) based on Fitzpatrick Skin Type (FST). [FST I vs. Type II vs. Type III-V].